A Mini-Review of Arabic Poetry Meter Classification

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***Abstract*—Poetry is an important art form in human history. Many poems have been written to share emotions, express pride or gratitude, declare or resolve conflicts, among others. Writing good poetry requires expertise. Understanding the structure of a poem helps poets better choose words and phrases that reflect their ideas. Identifying the meter of a poem helps in teaching new poets the different compositions they have at their disposal while also providing an easy way for professionals to detect broken poetry. This work investigates various efforts from the literature that tackled poetry meter classification, focusing on poetry written in Arabic. Different gaps in research have been identified including the need for proper analysis on the effects of diacritization on individual meters, the limited amount of research in spoken poetry due to scarce public datasets, and the need of further investigations regarding similar and rare poetry meters. We conclude with recommendations for dataset creation and modeling strategies to close these gaps and outline directions for future research in Arabic poetry meter classification.**

***Index Terms*—Arabic Poetry, Arudi Prosody, Poetry Meter Classification, Poetry Dataset, Quality Education, Audio-Based, Spoken Poetry, Rule-based, Neural-based.**

I. Introduction

Poetry is a form of traditional linguistic arts. The art of portraying and expressing emotions, beliefs and historical events through rhythmic compositions of words and phrases. Arabic Prosody is the science that studies the composition of poems in Arabic. It defines the construction rules and rhythmic sound that form verses and poems. Through the study of Arabic prosody, poets learn how to distinguish well formed poems from broken ones. This article looks through recent efforts in poetry meter classification. It focuses on poetry written following the Arud Science, the science defined by Al-Khalil bin Ahmad Al-Farahidi (AD 718-786) and is applied to Arabic, Indian, Turkish, and Persian poetry.

A. Background

In Arabic, a poem is written as a sequence of verses. Each verse is composed of two halves. The first half is labeled as *Sadr* while the second one is labeled as *Ajuz*. Poetic feet are a set of words that resemble sound cues which form rhythmic sounds. A poem's meter is identified based on different combinations of poetic feet. The basic feet are shown in Table I. For example, the basic form of the Basit meter is written as *"mustafilun failun mustafilun failun"*.

TABLE I
Standard poetic feet in Arud Science

Poetic Feet	Arudi Encoding
fau lun	0/ 0//
fa ilun	0// 0/
mutafa ilun	0// 0///
mufa ilatun	0/// 0//
mafa īlun	0/0/ 0//
fa ila tun	0/ 0// 0/
mus taf ilun	0// 0/ 0/
maf u latu	0/ 0/ 0/0

There are 16 meters in Arabic poetry. Each with a unique basic form. Some alterations could occur to the basic feet which result in different forms. As many alterations create variations to poetic meters, intersections between meters are formed where a variation of one meter could make it identical to that of another meter. The *Arudi Encoding* is a symbolic representation that is used to represent the voiced and unvoiced letters sequence in a verse. The sequence is composed of two symbols: a '/' is used to denote voiced letters and a '0' is used to denote unvoiced ones. To determine the meter of a verse, poets would start by writing this sequence. Then, the sequence is segmented such that each segment represents one of the poetic feet. Finally, the poetic feet sequence is used to find the meter of the verse. Various researchers tried to design and implement a computerized version of this method which will be covered in the following section.

B. Deep Neural Networks (DNN)

Deep neural models used for meter classification fall into two families: Long Short-Term Memory (LSTM) and the lighter Gated Recurrent Unit (GRU). They read verses step-by-step and keep a running memory of past context; their bidirectional variants (BiLSTM, BiGRU) read the text from both directions, which helps better link past and future context. Another model used for sequence classification tasks is BERT. It is a Transformer encoder model that uses an attention mechanism. It looks at all tokens simultaneously and benefits greatly from large-scale pretraining where various

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models found in the literature can be finetuned to different tasks. The performance of each model can be seen in Table II and Table III. In short, BERT-based methods provide the best results at the cost of high memory costs while BiLSTM and BiGRU providing competitive results when trained properly.

C. Objectives

Because of the importance of poetry in many cultures and the rapid development in natural language processing schemes, we want to find how this development has affected poetry. More importantly, we want to see how meter classification has changed over time. The objective of this paper is better illustrated in the following research questions.

- RQ1: How do rule-based approaches compare with statistical and neural models in terms of accuracy and interpretability?
- RQ2: What works and what does not for end-to-end spoken-poetry meter classification?
- RQ3: What are the key limitations and future research directions?

II. Literature Review

Over the years, linguistics and computer scientists have attempted to use computerized solutions to determine the meter of a poem. More recently, deep learning techniques have been utilized to analyze and predict a poem's meter. In this section, various poetry meter classification methods from the literature are presented. To understand the transition in classification methods, the different methods are divided into into groups based on the processes and classification algorithms being used. Figure 1 shows common standard workflows used for meter classification across the different methods.

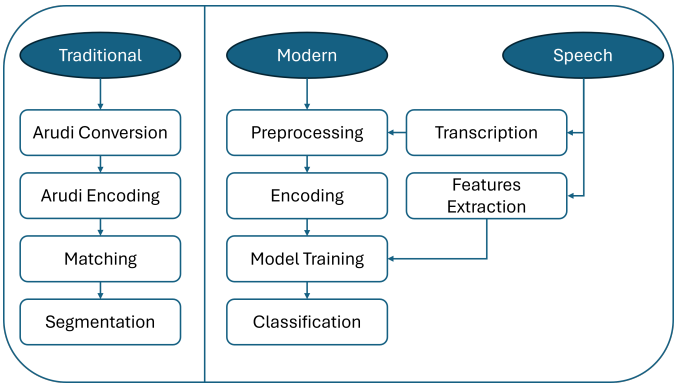


Fig. 1. Workflow of different meter classification methods

TABLE II
Summary of previous research. Type and size refer to the type and size of the data. {T: Text, S: Speech}. Text is counted by number of verses. Audio is counted by number of recordings

Ref.	Year	Type and Size	Category / Method
[1]	2013	T, 128	Traditional (CFG)
[2]	2016	T, not specified	Traditional (FSA)
[3]	2018	T, 417	Traditional (Sequence Matching)
[4]	2019	T, 150,000	Modern (RNN)
[5]	2021	T, 7,000	Modern (DT, SVM, ANN)
[6]	2024	T, 14,000	Modern (CAT-BiGRU)
[7]	2025	T, 183,909	Modern (BiLSTM, BiGRU)
[8]	2020	T, 55,440	Modern (ANN, LSTM, BiLSTM, GRU)
[9]	2025	T, 1,862,046	Modern (LSTM, BiLSTM, GRU)
[10]	2022	T, 1,628,368	Modern (BiLSTM)
[11]	2024	T, 2.09 million	Modern (BERT-Based model)
[12]	2020	S, 230	Audio (LPCC, MFCC, SVM)
[13]	2025	S, 403	Audio (MFCC, MTF, i-vector, SVM, DNN)
[14]	2023	S, 68	Audio (ML, TM, CNN)
[15]	2025	S, 3,936	Audio (Wave2Vec2, BiLSTM)

A. Traditional

We are using the word traditional to denote methods that incorporate the classical method of converting a verse to the arudi form and do not make use of machine or deep learning methods for classification.

Alnagdawi, Mohammad A., Hasan Rashideh, and F. Aburumman developed a tool that attempts to find the meter of a verse using Context-Free Grammar (CFG) [1]. Initial steps include rewriting the verse in the arudi form then segmenting it into short and long sound. For each meter, a set of rules are defined which map sound segments into poetic feet. The authors tested this approach with 128 verse. The tool achieved an accuracy of 75%. The authors noted that the tool requires fully diacritized verses. Dahab, Mohamed Y. et al. proposed an approach that utilizes Finite State Automata (FSA) to find the meter of a verse [2]. The authors defined conversion steps that generate a sequence of Ones and Zeros based on the input text. The sequence is matched to one of the poetic feet by following a FSA. To address alterations that could occur, the author defined various lookup tables for each case. If an input does not reach a final state, a sequence of backtracking and table lookups are used to find the correct altered poetic feet. A final FSA is used to find the verse's meter based on the poetic feet. Although the proposed approach seems promising, the authors did not include any evaluation data on the effectiveness or the accuracy of the proposed approach. In [3], Abuata, Belal and Al-Omari, Asma did also use a traditional pattern matching approach but only used the first half of the verse. The first step is to ensure that the verse is properly and fully diacritized; Though no diacritization rules or methods were

discussed. The seconds step is to rewrite the first hemstitch of the verse in the arudi form. By scanning the converted verse, the authors generate a sequence of symbols to denote short and long syllables. The sequence is then matched against sequences for each meter where a match is reported as the meter of the verse. The proposed method was testing on 417 verses where it correctly classified 82.2% of them. The authors did note that some verses do match multiple meters and the only way to address this is by scanning the whole poem.

B. Modern

A method is categorized as modern if it does not incorporate the conversion to arudi form or uses machine or deep learning techniques for classification.

In [4], Yousef, Waleed A. et al. tried three encoding methods. The first method is a basic one-hot encoding where each letter is represented by a sparse vector with a single 1 denoting the letter. Due to the inclusion of diacritics, this resulted in a large sparse vector. To remedy this, the author attempted a two-hot encoding method where the letter and the diacritic are encoded separately and are joined together afterwards. The third method is a binary encoding. Out of the three encoding methods, the authors were able to achieve the highest accuracy using the binary encoding with an accuracy of 96.38%. To test various machine learning models, Alqasemi, Fahd et al. in [5] proposed a curated set of features. The data used comes from [16]. To account for computational constraints, the authors randomly selected 10 of the 14 meters included in the dataset. Then, they randomly sampled 7000 verses from the included meters. A features matrix composed of letters' vectors and a cluster variable calculated using the K-means algorithm. The authors tested Decision Tree (DT), Support Vector Machines (SVM) and Artificial Neural Network (ANN). The tested models achieved accuracies between 95.45% and 96.34%. In [6], Al-shathry, Najla et al. developed a model for arabic poetry meter classification. The proposed model is based on CAT-BiGRU, an enhanced deep learning model that incorporates convolutional layers, Attention mechanisms, and a BiGRU model. The model's parameters where fine-tuned using ASCA-PSO, an adaptive hyperparameter tuning algorithm that combines the sine-cosine algorithm with swarm particle optimization to better optimize deep learning models. The authors used the same dataset from [16]. The proposed model was able to achieve an average accuracy 98.53% while requiring considerably less computational time compared to other tested models. Alqudsi, Zaid and Bilal, Muhammad and Wasly, Saud investigated the effects of epochs count, diacritics and data imbalance in poetry meter classification using multiple deep learning models [7]. A curated subset of the Arabic Poem Comprehensive Dataset 2 (APCD2) dataset [10] was used. By experimenting with a BiLSTM model, the authors noted that the little improvement is achieved when using more than 20 epochs. The difference between a balanced set of 165k verses from 11 meters and an imbalanced set of 183k verses from 16 meters were very minor while the difference between using diacritics or not was noticeable where removing the diacritics caused a decrease in accuracy from 96.57% to 94.26%. The authors tested two BiLSTM models and one BiGRU where the accuracy of each model was 96.57%, 96.38% and 94.76% respectively. M. S. Al-shaibani, et al, attempted a novel approach in Arabic poetry meter classification where they removed the diacritics from the poetry [8]. The dataset used contains 55,440 verses and covers 14 of the Arabic poetry meters. The authors tested multiple architectures including basic neural networks, LSTMs, GRUs and BiLSTMs. They also tested various network sizes and were able to achieve an accuracy of 94.32% on an independent test set when using a 5 layer BiGRU network. Mutawa, A. M., and Ayshah Alrumaih evaluated various deep learning models in poem meter classification [9]. The models used are GRU, LSTM, and BiLSTM. To train the models, the authors used the Arabic Poem Comprehensive Dataset (APCD) [4]. The authors experimented with training the model with full-verses and half-verses where the right and left verses are processed separately. According to the authors, their solution was able to achieve an accuracy of 97.53% when using full-verses with a BiLSTM model, and 95.23% accuracy when using half-verses with the BiLSTM model. Additionally, the authors indicated that the number of hidden layers affects the performance of the model as their model used only 3 hidden layers while the model from the related works used 4, 5, and 7 hidden layers. Abandah, Gheith A., et al. have used a BiLSTM model for poem meter classification [10]. The authors enhanced and extended the dataset from [4]. They removed some of the records from the old set, the ones with a missing half or are too short/long. Additionally, they added some prose to the dataset to facilitate training the model in distinguishing between prose and poems. The model used by the authors is a BiLSTM recurrent neural network. Many experiments were carried out including hyper-parameter tuning, model architecture modifications, utilizing multiple verses from a poem to improve the prediction accuracy, and the inclusion and exclusion of diacritics in training. Large language models such as BERT [17] have proven to be capable of achieving high performance across various natural language processing tasks. To leverage the power of this model, many researchers are developing BERT-based models. Qarah, Faisal. introduced a new BERT-based model trained on Arabic poems [11]. To train the model, the author used AraPoems dataset [18]. This dataset is publicly available and contains over 2 million verses with various information such as their meter, rhyme and poet. The authors tested their model in a multitude of tasks including meter classification and sentiment analysis. The model was able to achieve an overall accuracy of 0.9903 in classical meters classification. Both the model and the dataset are publicly available.

C. Speech-Based

Most of the efforts from literature use text-based approaches for poem meter classification due to the abundance of resources needed for model training and evaluation. Al-Talabani, Abdulbasit K. [12] experiments with meter classification

through LSTM using speech instead of text. The data used to train the model is prepared by the author and includes 230 full verses spoken by 2 speakers and only covers 3 Arabic poetry meters. To extract the features from the input data, the author used LPCC and MFCC separately and experimented with combining them. Additionally, they evaluated speaker dependent and independent meter classification and used an SVM classifier as a baseline. The results of the experiment might not be comparable to other attempts because the input data are relatively small, only covering 3 meters compared to the conventional 14-16, and the type of data being speech compared to the commonly used text format. Rajeev Rajan, et al. have used a fusion of MFCCs, i-vector, and MTF in their experiment in poem meter classification for the Indian language [13]. The dataset used was custom made and includes 403 audio files and covers 6 poem meters. The authors compared using a combination of MFCC and MTF against i-vector and MTF with an SVM and a DNN classifiers. According to the authors, using i-vector combined with MTF increases the accuracy from 80.83% to 86.66% while using a DNN for classification. Another attempt at meter classification from speech data was made by Nahar, Khalid M.O., et al, [14] where they prepared a dataset of verses collected from popular websites. The total number of verses collected is 68 which is comparably small to the number of verses found in [12]. The authors compared three methods for classification; Popular machine learning methods were used including Random Forests, Naïve Bayes and KNN. The second method utilizes a frequency plot of the sound signals and uses template matching to calculate the similarity between each verse's plot with that of each poetry meter. The third method used is Deep Learning where the authors used the VGG16 CNN model for prediction. Using a CNN model with a relatively small number of samples proved ineffective as the accuracy of the model was reported as 24.8%. The last attempt at audio-based poetry meter classification was carried out by Al-Shaibani, Maged S., Zaid Alyafeai, and Irfan Ahmad. [15]. The authors used a private dataset of audio recordings to train their model alongside a publicly available dataset to test the generalization of the proposed approach. The first dataset is composed of 3,668 audio recordings while the second one is composed of only 268. Two approaches were proposed by the authors. The first approach is an end-to-end system where it takes audio files as input and gives the predicted meter as output. The second approach uses a speech to text transcription model to facilitate the use of text-based meter classification models. The end-to-end system gave mediocre results with an overall accuracy of 40%. The authors suggested that the number of samples was not sufficient to properly train the model. The transcription based model was able to achieve an accuracy of 89% on the first dataset and 95% on the second one. As noted by the authors, important factors to the classification accuracy were the accuracy of the transcription model and the similarity between poetry meters. Errors caused by the transcription model propagate further into the system causing the classification module to fail. The classification module could also misclassify some samples due to the similarity between various poetic meters, such as Mujtath and Mudari.

III. Discussion

After analyzing various efforts from the literature, several points can be highlighted regarding poetry meter classification. First is the shift from traditional rule-based approaches to more modern statistical and neural based ones. Other than the rise of language models and ease of access to powerful machine learning tools, it is possible that a factor in the shift is the size and complexity in rule-based systems inherited by the complexity of the medium they are being developed for.

A. Rule-based systems against Statistical or Neural models

As mentioned before, the complicity of rules associated with poetry makes it difficult to define a conclusive set of rules that handle all cases. Additionally, the reliance on diacrtized verses limits the amount of data available for testing. As such, many modern approaches achieve considerably better results compared to traditional ones because they do not rely on the state of diacritization and the abundance of usable data.
Rule-based system were able to achieve and accuracy of 75% [1] and 82.2% [3] though both did not use publicly available dataset to allow for comparability. Most modern efforts have used publicly available dataset including the following:

- MetRec [16]
- APCD [19]
- APCD2 [18]

The accuracy of statistical and neural models exceed that of rule-based ones. Maged S. Al-shaibani, Zaid Alyafeai, Irfan Ahmad [8] were able to acheive an accuracy of 94.32% when using a BiGRU model and testing it with undiacritized verses. Qarah, Faisal [11] was able to achieve an accuracy of 99.03% with a BERT-based model. Abandah, Gheith A. et al. [10] were able to achieve an accuracy of 97.27% with a BiLSTM model.

B. Meter classification from spoken poetry

The ease of access to audio data opened the gates to improvements to speech processing tools and techniques. Though using those techniques for poetry meter classification is yet to be properly utilized. From a prosodic standpoint, the most informative acoustic cues are rhythmic periodicity and phrase-level pitch movement, with timbral and spectral cues [13]. A representative, language-agnostic pipeline fuses musical texture features rhythmic, melodic, and timbral with speech features and a variability-robust embedding. This setup resulted in a classification accuracy of 86.66% with a DNN classifier. Given the demonstrated language-agnostic nature of the MTF+i-vector approach, this method could be used for Arabic poetry where meters likewise arise from long/short syllabic timing. A major obstacle for spoken poetry processing is the lack of large public datasets. As noted in [15], the proposed end-to-end model was severely affected by the small number of data even with as much as 3,936 recordings. Other

efforts have had considerably smaller datasets which limits the comparability especially that most of the data are not public.

C. Research Gaps

In an effort to guide future research and highlight possible improvements to existing works, various authors suggesting the following:

1) Properly investigating the effects of diacritics in meter classification
   Traditional methods rely on diacritics to work. As such, many have discussed experimenting with ways to eliminate the need of diacritics or develop an automatic diacritization tool. The authors in [4] [5] [7] [10] have all attempted to include diacritized verses in their data. Since all public datasets come with a mix of diacritized, semi-diacritized and undiacritized verses, the results obtained show limited improvements when diacritics are included. In [11], the author was able to achieve the highest recorded class accuracy in multiple meters without using diacritics. The effects of diacritics in the classification results needs to be further studied including how it affects the results of individual classes.
2) Finding effective ways to solve the confusion between similar meters and low accuracy of classifying rare meters
   Due to the interpretability limitations introduced by deep learning techniques, the confusion between similar meters is hard to solve. The confusion comes from the different variations of meters. As traditional works utilize pattern or sequence matching techniques, they attempt to handle the situation where a verse matches multiple similar meters by processing more verses of the poem with the goal of finding one that matches the actual meter as mentioned in [1].
3) With meter classification of spoken poetry gaining attention, the intersection between different domain could prove useful for finding better better and more suitable solutions to this problem. Since poetry in Arabic, Urdu, Persian, and Turkish use syllabic and rhythmic features, a thorough study on speech signal processing, poetry composition, and linguistic properties could highlight the important features needed for accurate, language-agnostic meter classification.
4) Preparing a public dataset of spoken poetry records
   there is a limited number of published research in audio-based poetry meter classification. This could be caused by the lack of public datasets. The abundance of speech recognition technologies provides a way to utilize recorded poetry. While recent efforts have shown incredible results, it usually handles poetry similarly to how prose are being handled. With poetry being a rhythmic sound-based art, it would seem logical that such features would allow for better results when incorporated in the classification process.

IV. Conclusion

Writing good poetry takes time and effort. By better understanding the concepts of the medium, new poets should be able write better poems easily. By developing tools to help analyze poetry patterns, identify issues and give feedback to the user, it is possible that more people will be encourage to learn how to write poetry and experience the joy of expression through this traditional form of art. The literature shows a clear shift from rule-based systems to statistical and neural methods for poetry meter classification, driven by the difficulty of crafting comprehensive rules and the dependence on fully diacritized text. Rule-based systems have not been tested on public corpora which limits the comparability to other methods. Across public corpora (MetRec; APCD/APCD2), great results are achieved by machine-learning models, and BiGRU-, BiLSTM-, and BERT-based variants remain robust even without diacritics. Spoken meter classification remains nascent, constrained by small, often non-public audio datasets that limit generalization and comparability. Key gaps include (i) rigorous quantification of diacritic effects; (ii) mitigation of confusions among metrically similar classes and improved recognition of rare meters; and (iii) construction of large, standardized, publicly available speech datasets that capture rhythmic and prosodic cues. Addressing these gaps should yield more reliable, reproducible, and speech-aware meter classifiers.

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TABLE III
METHODS, FEATURES AND METRICS FOR ARABIC (AND RELATED) POETIC METER CLASSIFICATION

Citation	Method (Type)/Availability	Features / Inputs	Dataset/Availability	Metrics	Best Score
[1]	Rule-based	Context-Free Grammar over tafāīl templates; syllabic mapping	128 Arabic verses (source NR)	Acc.	75%
[3]	Rule-based	Deterministic tafāīl grammar; orthographic normalization	417 Arabic verses (source NR)	Acc.	82.2%
[4]	Neural (LSTM)	Char-seq; optional diacritics	$\sim$1.7M verses (web) 16 meters	Acc.	96.38%
[8]	Neural (BiGRU)	Char-level; no mandatory diacritization	MetRec: 55,400 verses 14 meters	Acc.	94.32%
[7]	Classical ML (SVM, etc.)	Customized feature extraction: orthographic/prosodic cues	3530 poems (Arabic)	Acc., F1	96.57%
[10]	Neural (LSTM)	Joint diacritization + meter classification; char-seq	Yousef et al. corpus (+ prose)	F1	97.27%
[9]	Neural (DL)	Deep models; performance analysis across params	Classical Arabic poems	Acc., F1	97.53%
[11]	Transformer (BERT-like)	Pretrained Arabic poem BERT; token/char variants	Arabic poem corpora	Acc., F1	99.03%
[6]	Hybrid meta-heuristic (ASCA) + CNN	VGG16 embeddings tuned by Adaptive Sine Cosine Alg.	Arabic poem corpus	Acc.	98.62%
[12]	Neural (LSTM) + SVM (baseline)	MFCC / acoustic features	230 verses (recited) 3 meters	Acc.	(SD 99.57%, SI 88.89%)
[13]	i-vector + MTF (fusion)	Spectral + musical tempo features	Malayalam (POMET) 8 meters	Acc., F1	86.77%
[14]	ML + Template + Deep (comp.)	Tafāīl template matching, ML baselines, deep models	Arabic classical poems (compiled)	Acc., F1	92.2%
[15]	Hybrid: Wav2Vec2 (ASR) $\rightarrow$ Text LSTM	ASR transcripts + char-seq text classifier; LM rescoring	2 recitation sets: Baseline, Benchmark (new)	Acc., P./R./F1	BL 87%, BM 92%

Notes: Acc.=accuracy; P.=precision; R.=recall; SI=speaker independent; SD=speaker dependent; BL=Baseline dataset; BM=Benchmar dataset; NR=Not reported

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